



End Term (Even) Semester Regular/Back Examination May 2026

Name of the Program and semester: B.TECH II SEM
Name of the Course: Programming for Problem Solving
Course Code: TCS 201
Time: 3 hours

Roll no. 2519216

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question are 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q.1	(20 Marks)	
a)	Assuming that the given segments of C and python programs does not prompt any compile time and/or run time errors, predict the output for them, when the following segments of code are executed on a 32-bit machine. 1. <pre>#include <stdio.h> void main() { int a = 6, b = 15; int *p = &a; int *q = &b; p = q; *p = *p + *q; *q = *p-*q; printf("%d %d", a, b); }</pre> 2. <pre>#include <stdio.h> void main() { char str[]="Quick brown fox"; char *cstr=str; printf("%c ",*(cstr++)); printf("%c ", *cstr); printf("%c", *(cstr+1)); }</pre>	CO1 CO3 CO5 CO6
	3. <pre>#include <stdio.h> int main() { int a[3][3]={0,1,1,1,0,1,1,1,0},i,j; for(i=0;i<3;i++) { for(j=0;j<3;j++) { a[i][j]=!a[i][j]; } } printf("%d", a[0][0]+a[1][1]+a[2][2]); return 0; }</pre> 4. <pre>#include<stdio.h> int main() { FILE *fp=fopen("data.txt","w+"); char buf[]="It was raining cats and dogs"; fputs(buf,fp); rewind(fp); fseek(fp,-4,2); fscanf(fp,"%s",&buf); printf("%s", buf); }</pre>	
	v) <pre>x="World" y="Weird" z="Weird" x, y = y+" "+ z , x print(x , y)</pre>	
b)	Draw a flowchart to read a string str and store in the memory allocated at run time using DMA. Pass this string into a function named <i>Encrypted_str()</i> which will encrypt string by replacing all the with the digits by '@' and all the vowels in lower case by '&'.	CO2



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c)	Draw a flowchart to read a string. Swap its first and last vowel and display the final string.	CO2				
	<table><tr><td>Sample Input:</td><td>Sample Output:</td></tr><tr><td>Inputted string: how are you?</td><td>Final string: huw are yoo?</td></tr></table>	Sample Input:	Sample Output:	Inputted string: how are you?	Final string: huw are yoo?	
Sample Input:	Sample Output:					
Inputted string: how are you?	Final string: huw are yoo?					
Q.2						
	Write a 'C' program to input a matrix of size $m \times n$ and print sum of all the boundry elements. (20 Marks)	CO1				
a)	<table><tr><td>Sample Input:</td><td>Sample Output:</td></tr><tr><td>Row size: 3 Col size: 4 Elements of matrix: 1 2 3 4 4 5 6 7 6 5 4 9</td><td>Sum: 46 (1 + 2 + 3 + 4 + 7 + 9 + 4 + 5 + 6 + 4 + 1)</td></tr></table>	Sample Input:	Sample Output:	Row size: 3 Col size: 4 Elements of matrix: 1 2 3 4 4 5 6 7 6 5 4 9	Sum: 46 (1 + 2 + 3 + 4 + 7 + 9 + 4 + 5 + 6 + 4 + 1)	
Sample Input:	Sample Output:					
Row size: 3 Col size: 4 Elements of matrix: 1 2 3 4 4 5 6 7 6 5 4 9	Sum: 46 (1 + 2 + 3 + 4 + 7 + 9 + 4 + 5 + 6 + 4 + 1)					
b)	State the purpose of Dynamic Memory Allocation (DMA) in 'C'. What are the various methods to achieve DMA? Explain with suitable syntax and example. Which header file must be included to use these functions? When does a memory leak arise? What precaution must a programmer practice to avoid it?	CO3				
c)	Develop a 'C' program to read n integer numbers and store in the array created using dynamic memory allocation. Replace each element of array with a new element in such a way that the element become completely divisible by 8. Print the final array.	CO3				
	<table><tr><td>Sample Input:</td><td>Sample Output:</td></tr><tr><td>Value of n: 5 Elements of array: 16 22 41 60 75</td><td>Final array after changes: 16 16 40 56 72</td></tr></table>	Sample Input:	Sample Output:	Value of n : 5 Elements of array: 16 22 41 60 75	Final array after changes: 16 16 40 56 72	
Sample Input:	Sample Output:					
Value of n : 5 Elements of array: 16 22 41 60 75	Final array after changes: 16 16 40 56 72					
Q.3						
	(20 Marks)					
a)	Describe structure? What is the purpose of dot and arrow operator in structure. Explain with suitable example. Also describe the following terms with suitable example. i) self-referential structure ii) structure v/s union iii) pointer to structure	CO4				
b)	Management of a factory wants to fetch the detail of working hours of their worker on a particular day. Help management team with a 'C' code which will input detail of n workers of the factory. Display their complete detail and also find working hour of each worker. Note: Create a structure <i>Worker</i> with the data members <i>W_name(char type)</i> , <i>W_id(int type)</i> , <i>PunchInTime(int int int)</i> , <i>PunchOutTime(int int int)</i> . (consider the format: <i>hr min sec</i> for <i>PunchInTime</i> and <i>PunchOutTime</i>).	CO4				
c)	Create a structure <i>Book</i> having data members <i>B_name(char type)</i> , <i>B_author(char type)</i> , <i>B_sellingcount(int type)</i> and <i>B_price(float type)</i> . Develop a 'C' code to read detail of n books. Sort the books on basis of their selling count in descending order. Also print the detail of book having highest and lowest selling count.	CO4				
Q.4						
	(20 Marks)					
a)	Explain the need of file in 'C' programming. What do you mean by sequential access and random access? How we can achieve random file access in C? Explain in detail with suitable syntax and uses.	CO5				
b)	Write a menu-driven program using dynamic memory allocation for the following operations: 1. Allocate memory for n integers 2. Count how many numbers are palindrome	CO3				



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	3. Display elements 4. Resize the array for 5 more elements 5. Free memory 6. Exit					
c)	Develop a 'C' code to write n integer numbers into a file named "FILE_1". Read the file and copy all the three-digit number to another file named "FILE_2" in reverse sequence. <table><tr><td>Sample Input</td><td>Output Input</td></tr><tr><td>Number stored in FILE_1: 23 532 2 783 45 12 986 6</td><td>Number stored in FILE_1: 986 783 532</td></tr></table>	Sample Input	Output Input	Number stored in FILE_1: 23 532 2 783 45 12 986 6	Number stored in FILE_1: 986 783 532	CO5
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Number stored in FILE_1: 23 532 2 783 45 12 986 6	Number stored in FILE_1: 986 783 532					
Q.5	(20 Marks)					
a)	"Python is a popular programming language". Justify your answer. How it is different from 'C' language? Explain various data types available in python with proper example. Also explain any two categories of operators available in Python.	CO6				
b)	Develop a menu driven Python code which use multiple functions for the following menus: 1) $f1()$ accept four-digit integer and display sum of its first and last digit 2) $f2()$ accept an integer number and print its factors.	CO6				
c)	Develop a python code which accept an integer number and check whether number is Strong Number or not. Note: Example of Perfect number: 145 ($1! + 4! + 5! = 145$)	CO6				